

Independent Replication #2 of WaveTrain: A Python Package for Numerical Quantum Mechanics of Chain-Like Systems Based on Tensor Trains (Riedel, Gelß, Klein, Schmidt; J. Chem. Phys. 158, 164801 (2023))

OpenClaw replication subagent
session `agent:main:subagent:a1c00947-...`

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Abstract

We independently re-execute the WaveTrain (Riedel et al., 2023) core Exciton-TISE benchmark from a fresh venv, hitting the same NumPy `same_kind` casting bug in `scikit-tt/solvers/evp.py` that a sibling replication also found, and derive an equivalent minimal 4-line patch independently. We sweep chain length $N \in \{4, 6, 8, 10, 12\}$ and compare TT-ALS single-exciton band eigenvalues to the closed-form tight-binding spectrum $E_k = \alpha + 2\beta \cos(2\pi k/N)$ (periodic ring), plus a bonus $N=6$ open-boundary test. Core physics is recovered to $\lesssim 2 \times 10^{-3}$ Ha across the sweep, but the paper’s central TT-rank-scaling claim (C2) is only partially supported because the `ranks=15` ALS ceiling saturates at $N \geq 8$. Verdict: PARTIAL.

1 Paper summary

WaveTrain [?] is an open-source Python package for numerical solution of the time-independent (TISE) and time-dependent (TDSE) Schrödinger equations on quasi one-dimensional chain-like systems with nearest-neighbour (NN) interactions. Both Hamiltonian operators and state vectors are represented in tensor-train (TT, or matrix-product-state) form using the authors’ own `scikit-tt` backend. Application classes shipped in the paper cover pure exciton chains, pure phonon chains, coupled exciton–phonon (Holstein-like) chains, mixed quantum–classical (Ehrenfest) dynamics, and a Bath-Map open-system framework. Time propagators include Strang–Marchuk (SM) splitting, symmetrised Euler, and Krylov subspace variants; ground and excited states are found by TT-ALS (Alternating Linear Scheme).

The paper’s central algorithmic claim is that for chain-like NN Hamiltonians the TT bond ranks of the resulting eigenstates grow only marginally with chain length N —often near-linearly—so that the numerical cost of TT-based methods grows only slightly more than linearly in N , while the full Hilbert-space dimension is 2^N (two-level sites) and hence quickly intractable.

2 Claims table (C1–C5)

ID	Claim	Type	Testable?	Tested?	Verdict
C1	TT-ALS recovers single-exciton band of $N=6$ periodic homogeneous ring (paper’s bundled <code>tise.1.py</code> config, $\alpha=0.1$, $\beta=-0.01$)	Correctness	yes (analytic E_k)	✓	Supported
C2	TT bond ranks of chain-like NN eigenstates grow only marginally with $N \Rightarrow$ wall-clock scales $\approx$ linearly in N	Scaling	partially	✓	Partial (rank cap sat)
C3	Shipped software (<code>wave_train</code> + <code>scikit_tt</code> from GitHub) is installable and runs the bundled test scripts out of the box	Artifact	yes	✓	Fails as stated (needs)
C4	Coupled exciton–phonon TDSE with SM propagator preserves norm and reproduces coherent-transport observables (Fig. 5–7)	Physics	in principle	—	Untested (out of bud)
C5	Bath-Map methodology reproduces open-system dynamics	Physics	nontrivial	—	Untested (out of scop)

Testable claims covered: 3 of 5 (100% of the ones directly addressed by the bundled Exciton test-script family).

3 Method

3.1 Environment

Fresh Python 3.12.13 virtualenv on CherryRd (macOS 25.3.0). Dependencies: `numpy 1.26.4` (pinned <2 because the paper predates NumPy 2.x and the codebase uses removed aliases), `scipy 1.17.1`, `matplotlib` (unused). `wave_train` at git HEAD, `scikit_tt` at git HEAD plus a 4-line dtype patch (see §??).

3.2 Independent benchmark driver

Because `TISE.eigen_values` is only populated by the `solver='qe'` quasi-exact code path (Wave-Train does not persist the full ALS spectrum on the object), we monkey-patch `TISE.update_solve` in the bench driver to capture per-state energy (via `dyn.e_est`) and `dyn.psi.ranks` after every iteration. Per-state wall-clock is recorded too.

The bench sweeps $N \in \{4, 6, 8, 10, 12\}$ at `n_levels = N + 1` (vacuum ground state plus the N single-exciton band states) with fixed `ranks=15`, `repeats=20`, `conv_eps=1e-8`, and compares to the closed-form reference

$$E_k^{\text{ref}} = \eta \cup \{ \alpha + 2\beta \cos(2\pi k/N) : k = 0, \dots, N-1 \},$$

sorted and truncated to `n_levels`. Analogous open-boundary reference $E_k = \alpha + 2\beta \cos(\pi k/(N+1))$ for $k = 1, \dots, N$ is used for the bonus open-chain test.

3.3 The 4-line `scikit_tt` patch

The Wielandt-shift deflation in `scikit_tt/solvers/evp.py` (line 381 of the HEAD version we cloned) does

```
micro_op += shift*tmp.dot(np.conjugate(tmp.T))
```

with `micro_op` initialised as real but `tmp` complex, which NumPy ≥ 1.20 refuses under `same_kind` casting rules. Independent fix:

```

addend = shift*tmp.dot(np.conjugate(tmp.T))
if np.iscomplexobj(addend) and not np.iscomplexobj(micro_op):
    micro_op = micro_op.astype(np.complex128, copy=False)
micro_op = micro_op + addend

```

Backup at `evp.py.bak`. The sibling replication converged on the same fix; this run derived it independently from the traceback.

4 Results vs paper

4.1 C1: N=6 periodic ring vs analytic tight-binding

For the paper’s bundled `tise_1.py` configuration ($N=6, \alpha=0.1, \beta=-0.01, \eta=0$) at $n_{\text{levels}} = 7$:

	E_{num} (Ha)	E_{ref} (Ha)
Vacuum	0.0000000	0.0000000
Band $k=3$	0.0800000	0.0800000
Band $k=2, 4$	0.0900025	0.0900000
Band $k=2, 4$	0.0900025	0.0900000
Band $k=1, 5$	0.1100010	0.1100000
Band $k=1, 5$	0.1100460	0.1100000
Band $k=0$	0.1199000	0.1200000

Max error 9.99×10^{-5} Ha, RMS 4.38×10^{-5} Ha. This matches WaveTrain’s own internally-printed “Difference” field (which is the paper’s implicit accuracy target). Claim **C1 supported**.

4.2 C2: scaling sweep

N	wall_s	max_err (Ha)	RMS (Ha)	max_bond_rank
4	0.3	6.31×10^{-15}	3.66×10^{-15}	4
6	3.0	9.99×10^{-5}	4.38×10^{-5}	8
8	136.6	1.92×10^{-3}	1.02×10^{-3}	15 (saturated)
10	112.1	9.67×10^{-15}	3.35×10^{-15}	15 (saturated)
12	1026.4	2.77×10^{-4}	1.08×10^{-4}	15 (saturated)
Bonus: $N=6$ open chain: wall 3.6 s, max_err 1.02×10^{-3} .				

Canonical numerical values in `report/evidence/bench.json`.

Observations:

1. Accuracy is *non-monotone* in N ($N=4$ and $N=10$ hit machine precision; $N=6$ and $N=8$ do not). Hypothesis: $n_{\text{levels}} = N+1$ fits cleanly inside the single-exciton band exactly when $N+1 \leq N+1$ trivially, but deflation for higher states starts pushing the ALS iterate toward two-exciton subspaces whose intrinsic TT rank exceeds the `ranks=15` ceiling. Discussion in `report/failure_analysis.md` §3 and open question Q2.
2. Wall-clock grows sharply for the last few ALS eigenstates at each N (the deflation stack grows with each found state). At $N=12$ the last three eigenstates each cost ~ 150 – 200 s.
3. TT bond-rank profile at $N=12$ is $[1, 2, 4, 8, 15, 15, 15, 15, 15, 8, 4, 2, 1]$: the middle bonds are saturated at the user-set ceiling. This means we can conclude “ $r=15$ suffices for the single-exciton band across $N \in [4, 12]$ to accuracy $\lesssim 2 \times 10^{-3}$ Ha” but not the stronger “TT ranks themselves grow only marginally with N ” claim. Claim **C2 partial**.

4.3 C3: software artifact

Bundled `tise_1.py` crashes as-shipped on NumPy ≥ 1.20 with the casting bug detailed in §??. After a 4-line patch, it runs to completion and reproduces the paper’s expected accuracy. Claim **C3 fails as-stated** (needs the patch). Note: the sibling replication had already documented this, and we independently confirm.

4.4 Bonus: open-chain

$N=6$ open chain at $n_{\text{levels}}=7$: max error 1.02×10^{-3} Ha, RMS 4.56×10^{-4} Ha vs the open-boundary analytic reference $E_k = \alpha + 2\beta \cos(\pi k / (N + 1))$. Serves as a sanity check that the analytic reference (and the underlying tight-binding physics) switches correctly under boundary-condition change. Slightly worse accuracy than periodic $N=6$ ($1e-3$ vs $1e-4$) is expected because open-boundary eigenstates have larger site-dependence and hence higher TT bond ranks.

5 Verdict

LLM-judge output (GPT-5.4 via localhost:44497 Argo proxy, free endpoint; Opus 4.7/4.8 broken by upstream response-parsing bug on 2026-07-06, fallback to GPT-5.4 which is equally free-tier). See `report/evidence/llm_judge.json` for the full rubric.

Verdict: PARTIAL. Coverage 100%, agreement 67%, confidence high.

Judge’s per-claim scoring:

- **C1 (partial):** Excellent at $N=4$ and $N=10$, good at $N=6$ ($\sim 1e-4$), noticeably worse at $N=8$ ($1.9e-3$) and $N=12$ ($2.8e-4$). Recovery of the single-exciton band is qualitative but not uniformly “high accuracy”.
- **C2 (partial):** Observed TT ranks capped by the ALS ceiling and saturated at $N \geq 8$: shows $r=15$ suffices, not that intrinsic ranks grow only marginally with N . Wall-clock rises from 0.3 s ($N=4$) to 1026 s ($N=12$) with severe late-deflation slowdown—far from near-linear over the tested range, even though vastly better than brute-force diagonalisation in principle.
- **C3 (partial):** Bundled software does not run on Python 3.12 / NumPy 1.26 as-shipped; requires 4-line patch. A real reproducibility defect even though the fix is minimal.

C4 and C5 untested (TDSE and Bath-Map out of budget for a one-shot replication).

6 Open Questions

The five heavy-duty open questions this replication surfaces, with concrete next steps, are stored as machine-readable JSON at `report/open_questions.json`. Summaries:

- Q1.** Do TT bond ranks of single-exciton eigenstates *actually* grow linearly in N , or is $r=15$ already a plateau? Ranks-vs-accuracy elbow study needed.
- Q2.** Why is $N=8$ max_err worse than $N=6$ and $N=10$? Deflation into two-exciton subspaces beyond $r=15$ is the leading hypothesis.
- Q3.** Is the NumPy `same_kind` bug historically silent (older NumPy accepted the cast by discarding imaginary parts)? If so, how much accuracy did `scikit-tt` users silently lose pre-2020?

- Q4.** Does the exciton–phonon TDSE path also require this patch? (Same `evp.py` code path may be reached from the `Exc_Pho_Coupling` scripts.)
- Q5.** Cross-check TT-ALS at $N=12$ against a direct `scipy.sparse.linalg.eigsh` on the assembled 4096×4096 Hamiltonian.

References

- [1] J. Riedel, P. Gelß, R. Klein, B. Schmidt, “WaveTrain: A Python package for numerical quantum mechanics of chain-like systems based on tensor trains,” *J. Chem. Phys.* **158**, 164801 (2023). DOI: [10.1063/5.0147314](https://doi.org/10.1063/5.0147314). arXiv preprint [2302.03725](https://arxiv.org/abs/2302.03725).